

STEP Interface Functional Specification

Institute of Technical Education – Data
Repository

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	21/11/2023	Initial version of ITE Data Repository IFS	AvePoint
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Document Acceptance Form

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specifications between the SaaS Skills Training Enhancement Portal (STEP) and Institute of Technical Education Data Lake for transferring all ITE raw data in STEP reporting DB.

STEP shall identify all ITE CET data with changes, generate corresponding CSV files for these data, deposit CSV files encrypted by PGP protocol in the SFTP server accordingly. New scheduled jobs to be implemented by ITE will retrieve, decrypt and process these PGP encrypted data files accordingly for downstream system needs.

1.2 Intended Audience

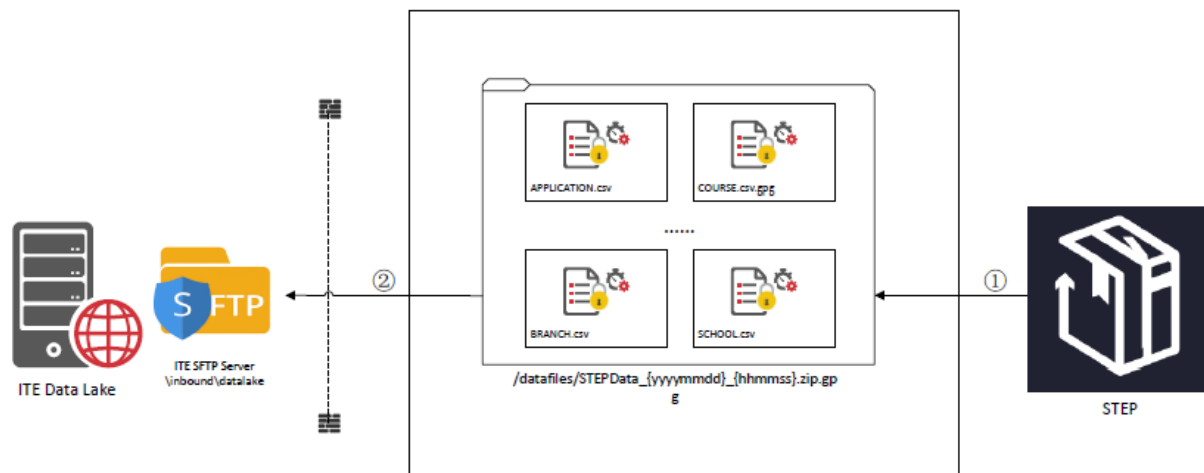
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both STEP and ITE Data Lake System.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **ITE Data Lake:** Institute of Technical Education Data Lake.
- **ITE SFTP Server:** A SFTP server that stores encrypted files used for data exchange.
- **STEP:** Skills Training Enhancement Portal System.



*The above system flow only shows some CSV files. For the interface file name of all ITE CET data CSV files, please refer to [2.2 Interface Design Summary](#).

SN	Description
1	STEP generates ITE delta data CSV files based on institution, and modified time on weekly basis. For file generation rules refer to 2.3. Business Rule
2	STEP push zip files under corresponding folder after encryption with PGP protocol: /({ENV})/Inbound/datalake/STEPData_{yyyymmdd}_{hhmmss}.zip.gpg ITE data lake system pulls the zip file from the SFTP server on demand.

2.2 Interface Design Summary

Interface File Name	STEPData_{yyyymmdd}_{hhmmss}.zip.gpg contains the following CSV files: Naming rule: DBTableName.csv, about 133 files in total, Refer to 2.4 Sample file – DB schema to find all table information. Checksum.txt Report_User.csv Report_School.csv Report_MyCourse.csv Report_Branch.csv
Data Source System	Skills Training Enhancement Portal
Data Destination System	Institute of Technical Education Data Lake
Method	SFTP
File Path	/{ENV}/inbound/datalake/ ENV: UAT, PROD
Encryption Method	PGP
Frequency	Weekly Sync
Delimiter in CSV	#
Folder Retention Policy	STEP generate zip file and push it under /{ENV}/inbound/datalake/ ITE is responsible for file retention since the SFTP is hosted on ITE side.
File Transfer Policy	STEP shall transfer empty CSV files with header info in zip file into ITE SFTP server if there are no new records added/updated in the table.
Possible Exception and Exception Handling	File loss or transfer failure due to STEP internal exception. E.g., Timer service down, etc. Exception handling steps: - STEP support team shall be informed by email of the transfer failure. - AP team checks the relevant logs to troubleshoot the cause of the failure. - AP redeposits the files in the SFTP folder.

2.3 Business Rule

CSV files generation rule

The ITE full data in STEP will be exported, containing but not limited to the following:

- Course,
- Application,
- Enrolment,
- Timetabling,
- Assessment,
- Graduation,
- Student Finance,
- Master Data.

Table record trigger point: First-time full load data sync + Subsequent delta data sync.

Delta Data sync trigger point: Delta Data sync mode is record level. For example, if the area of interest field in the course table is updated, send all the data fields of this record in the course table to the ITE CET data repository.

Filter condition:

- Institution = ITE,
- The data fields in the table have changed,
- Modified datetime later than last successful sync datetime.

Datetime field format: MM/DD/YYYY HH:MM:SS AM/PM (UTC+8), E.g., 11/14/2023 1:15:05 PM

Besides ITE only data, the global master data shared by all IHLs will be also included in the interface CSV files. E.g., Citizenship Type, Course Category, Course Type, etc.

Remarks: Student photos and supporting documents are not in the interface scope.

2.4 Sample Files

The following sample CSV files:



**ITE Data Repository
Sample Files.zip**

STEP DB schema:



**step-data-schema-20
23Dec.sql**

Checksum for all CSV files:



checksum.txt

3. Assumption

3.1 Assumption

No.	Assumption
1	The interface design is based on the current STEP DB structure, need to go through CR process when the change in STEP constitutes to a change in database or new fields are introduced which the original interface does not cover.